

A Regular-Class Existence Theorem for Santaló Gradient-Flow Curves

Run fa_banach_001, lane 19

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Source Question

Cordero-Erausquin, Fradelizi, and Langharst ask in Section 5.2 of arXiv:2409.05541 whether the time-dependent gradient-flow equation

$$s'(t) = -\frac{p}{2q} \nabla_z Q(t, s(t))$$

has at least one sufficiently smooth solution on $\mathbb{R}_+$ with $\lim_{t \rightarrow \infty} s(t) = 0$. Here $0 < p < 1$, $q = p/(p-1) < 0$, f_t is a heat or Fokker-Planck evolution, and

$$Q(t, z) = \log \int_{\mathbb{R}^n} \mathcal{L}_p(\tau_z f_t)(x) \, dx.$$

The source passage is shown in Figure 1.

This packet proves a positive answer for the regular class used in the source paper's core semigroup theorem: bounded compactly supported nonzero data under the Fokker-Planck flow. The arbitrary nonnegative-function case and the $p = 0$ volume-product endpoint are not claimed.

Remark 1 (Sign convention). *We use the first equality in the source's proposed equation, $s' = -(p/(2q)) \nabla_z Q(t, s)$, which is exactly the choice making the monotonicity condition used below hold with equality. With the source convention $\tau_a f(x) = f(x - a)$ one has $\nabla_z Q = q \mathbb{E}[X]$ under the tilted $\mathcal{L}_p(\tau_z f_t)$ density, so the equivalent expectation formula carries a minus sign. This packet is therefore aligned with the displayed gradient equation and with the monotonicity condition, not with the informal sign in that expectation rewrite.*

Definitions

Fix $0 < p < 1$ and $q = p/(p-1)$. For a nonnegative function g define

$$\mathcal{L}_p(g)(x) = \left(\int_{\mathbb{R}^n} g(y)^{1/p} e^{x \cdot y} \, dy \right)^q.$$

Let $P_t f$ be the Fokker-Planck flow as in the source paper and set $f_t = P_t f$. For $t > 0$ put

$$Q(t, z) = \log \int_{\mathbb{R}^n} \mathcal{L}_p(f_t)(x) e^{qz \cdot x} \, dx = \log \int_{\mathbb{R}^n} \mathcal{L}_p(\tau_z f_t)(x) \, dx.$$

For bounded compactly supported nonzero f , the source paper proves that Q is smooth on $(0, \infty) \times \mathbb{R}^n$, strictly convex in z , and finite for every z .

It is convenient to set

$$c_p = -\frac{p}{2q} = \frac{1-p}{2} > 0.$$

The source equation is therefore

$$s'(t) = c_p \nabla_z Q(t, s(t)). \quad (1)$$

Proof Intuition

Along the Fokker-Planck flow, any bounded compactly supported density converges exponentially, with derivatives on compact sets, to a centered Gaussian. For a centered Gaussian g , a direct computation gives

$$Q_g(z) = C + \frac{|z|^2}{2(1-p)}, \quad c_p \nabla Q_g(z) = \frac{1}{2}z.$$

Thus the requested ODE is, near $t = \infty$, a small nonautonomous perturbation of $s' = s/2$. A terminal-at-infinity solution can then be obtained from the integral equation

$$s(t) = - \int_t^\infty e^{(t-u)/2} R(u, s(u)) du,$$

where $R(t, z) = c_p \nabla Q(t, z) - z/2$ decays exponentially in t and is small Lipschitz in z on a large tail. The resulting small solution on $[T, \infty)$ is then extended backward by standard ODE continuation.

Auxiliary Estimates

Lemma 1 (Gaussian asymptotics of the vector field). *Let $f \geq 0$ be bounded, compactly supported, and not identically zero. Let $f_t = P_t f$. For every $R > 0$ there are constants $C_R, \lambda > 0$ and $T_R > 0$ such that, for all $t \geq T_R$ and $|z| \leq R$,*

$$\left| c_p \nabla_z Q(t, z) - \frac{1}{2}z \right| \leq C_R e^{-\lambda t},$$

and

$$\left\| D_z \left(c_p \nabla_z Q(t, z) - \frac{1}{2}z \right) \right\| \leq C_R e^{-\lambda t}.$$

Proof. The Fokker-Planck formula in the source paper can be written as a centered Gaussian factor multiplied by a Laplace transform of f at a parameter of size $O(e^{-t/2})$. Since f is bounded and compactly supported, that Laplace transform and all its derivatives admit a Taylor expansion at the origin with remainders $O(e^{-\lambda t})$ on compact x -sets. The same is therefore true for $f_t^{1/p}$ and for its Laplace transform $H_t(x) = \int f_t(y)^{1/p} e^{x \cdot y} dy$, again locally uniformly with derivatives.

Moreover the Gaussian upper and lower bounds in the source paper's class $\mathcal{C}$ are uniform for t large. Since $q < 0$, the functions $H_t(x)^{q e^{qz \cdot x}}$ are dominated, uniformly for $|z| \leq R$, by an integrable Gaussian. Differentiation under the integral sign is justified by the same domination, so $Q(t, \cdot)$ and its first two z -derivatives converge exponentially on $|z| \leq R$ to the corresponding quantities for the limiting centered Gaussian.

For $g = a\gamma_1$ one computes

$$\mathcal{L}_p(g)(x) = C_{a,p,n} \exp \left(-\frac{p^2}{2(1-p)} |x|^2 \right),$$

and hence

$$Q_g(z) = C'_{a,p,n} + \frac{|z|^2}{2(1-p)}.$$

Multiplying the gradient by $c_p = (1-p)/2$ gives the limiting vector field $z/2$. This proves the two displayed estimates. $\square$

Lemma 2 (No finite backward blow-up away from zero). *Let $0 < \varepsilon < T < \infty$. On $[\varepsilon, T] \times \mathbb{R}^n$ the vector field $F(t, z) = c_p \nabla_z Q(t, z)$ is smooth and has at most linear growth in z . Consequently every solution of $s' = F(t, s)$ defined at time T extends backward to all of $[\varepsilon, T]$.*

Proof. For $t \in [\varepsilon, T]$, the source paper's heat/Fokker-Planck regularization places f_t , its needed derivatives, and $\mathcal{L}_p(f_t)$ in the Gaussian class $\mathcal{C}$, with constants locally uniform in t . Thus

$$\mathcal{L}_p(f_t)(x) e^{qz \cdot x}$$

is squeezed between Gaussian tails with a linear tilt. Its normalized first moment is therefore bounded by $A_{\varepsilon, T}|z| + B_{\varepsilon, T}$. Since $\nabla_z Q(t, z)$ is q times that normalized first moment, F has at most linear growth. Standard Picard-Lindelof continuation then prevents finite-time blow-up on $[\varepsilon, T]$. $\square$

Main Partial Theorem

Theorem 1 (Regular Santaló gradient-flow curves exist). *Let $0 < p < 1$, $q = p/(p-1)$, and let $f : \mathbb{R}^n \rightarrow [0, \infty)$ be bounded, compactly supported, and not identically zero. Let $f_t = P_t f$ be its Fokker-Planck evolution and let Q be defined above. Then there exists a C^1 curve $s : (0, \infty) \rightarrow \mathbb{R}^n$ such that*

$$s'(t) = -\frac{p}{2q} \nabla_z Q(t, s(t))$$

for every $t > 0$, and

$$\lim_{t \rightarrow \infty} s(t) = 0.$$

In particular, this curve satisfies the source paper's condition (58) with equality, and hence belongs to the regular version of $\tilde{\mathcal{S}}_p(f)$.

Proof. Write

$$F(t, z) = c_p \nabla_z Q(t, z), \quad R(t, z) = F(t, z) - \frac{1}{2}z.$$

By Lemma 1, choose T so large that on the unit ball $B(0, 1)$ and for all $t \geq T$,

$$|R(t, 0)| \leq A e^{-\lambda t}, \quad |R(t, z) - R(t, w)| \leq \eta |z - w|,$$

where $\eta > 0$ is small enough that

$$\eta \int_0^\infty e^{-v/2} dv < \frac{1}{2}.$$

Increase T if necessary so that

$$A \int_T^\infty e^{(T-u)/2} e^{-\lambda u} du < \frac{1}{2}.$$

Let X_T be the closed set of continuous curves $s : [T, \infty) \rightarrow B(0, 1)$ such that $\lim_{t \rightarrow \infty} s(t) = 0$, equipped with the supremum norm. Define

$$(\Phi s)(t) = - \int_t^\infty e^{(t-u)/2} R(u, s(u)) du.$$

The estimates above show that Φ maps X_T into itself and is a contraction. Hence Φ has a unique fixed point s on $[T, \infty)$. The fixed point is C^1 by differentiating the integral equation, and it satisfies

$$s'(t) = \frac{1}{2}s(t) + R(t, s(t)) = F(t, s(t)).$$

Also $s(t) \rightarrow 0$ by construction.

It remains to extend the solution to all positive times. Lemma 2 extends the solution backward from T to every interval $[\varepsilon, T]$, $\varepsilon > 0$. These extensions agree on overlaps by uniqueness for the smooth vector field. Taking their union gives a C^1 solution on $(0, \infty)$ with the same terminal limit at infinity.

Finally, because $s' = -(p/(2q))\nabla_z Q(t, s(t))$, the expression in source condition (58) is

$$\frac{p}{2q}|\nabla_z Q(t, s(t))|^2 + \nabla_z Q(t, s(t)) \cdot s'(t) = 0.$$

Thus the curve is admissible in the regular $\tilde{\mathcal{S}}_p(f)$ sense. □

Scope and Novelty Check

The theorem proves the open-question existence statement for the source paper's regular bounded-compact-support class and the Fokker-Planck flow. It does not settle arbitrary nonnegative data, uniqueness among all terminal solutions, compact-region descriptions for all Santaló curves, or the endpoint $p = 0$ volume-product equation.

Cheap run indexes were searched for arXiv:2409.05541, Santaló, Nakamura, Tsuji, gradient flow, Laplace transform, and related convex-geometric terms. A bounded web search on July 19, 2026 found the source paper and open-problem summaries but no exact later answer to this gradient-flow existence question.

References

1. D. Cordero-Erausquin, M. Fradelizi, and D. Langharst, *On a Santaló point for Nakamura-Tsuji's Laplace transform inequality*, arXiv:2409.05541; Forum of Mathematics, Sigma 13 (2025), e125.
2. S. Nakamura and H. Tsuji, *The functional volume product under heat flow*, arXiv:2401.00427.

It would be interesting to know if, after translating f , the single point zero curve belongs to $S_p(f)$ or if the curves stay in a compact region that can be described. Actually, we don't even know if for the volume product, i.e. when $p = 0$, $t \mapsto \int f_t \int (f_t)^\circ$ increases when f or f° has barycenter 0, say. Note also that monotonicity ensures that $\int f_t \int (\tau_{s(t)} f)^\circ \leq (2\pi)^n$, and so $s(t)$ has to belong to the so-called 'Santaló region' [27, 16] of f_t .

For instance, in the trivial case where $f(x) = e^{-|x-a|^2/2}$ is a Gaussian centered at $a \in \mathbb{R}^n$, then it is readily checked that $s(t) = e^{-t/2}a$ is the only element of $S_p(f)$ for all $p \in [0, 1]$.

One can also look at the subset of curves in $S_p(f)$ that make our argument work with fixed $p \in (0, 1)$. In the proof of (32) in Theorem 3.1, given in Section 3.4, we see that establishing the monotonicity in t of $M_p(f_t)$ (assuming f belongs to the class $\mathcal{C}$ defined in (40)) amounted to the fact that

$$\begin{aligned} \alpha'(t) &= \partial_t Q(t, s_p(f_t)) + \nabla_z Q(t, s_p(f_t)) \cdot \partial_t s_p(f_t) \\ &\geq \frac{p}{2q} |\nabla_z Q(t, s_p(f_t))|^2 + \nabla_z Q(t, s_p(f_t)) \cdot \partial_t s_p(f_t) \\ &= 0, \end{aligned}$$

where $Q(t, z) = \log L(\mathcal{L}_p(f_t))(qz) = \log \int \mathcal{L}_p(\tau_z(f_t))$ is convex in z , and $\alpha(t) = Q(t, s_p(f_t))$. In particular, our choice of $z = s_p(f_t)$ ensured that

$$|\nabla_z Q(t, s_p(f_t))|^2 = 0 = \nabla_z Q(t, s_p(f_t)) \cdot \partial_t s_p(f_t).$$

But we see that we only need the curve to satisfy

$$\frac{p}{2q} |\nabla_z Q(t, s(t))|^2 + \nabla_z Q(t, s(t)) \cdot s'(t) \geq 0, \quad (58)$$

which leads to the subset $\tilde{S}_p(t) \subset S_p(f)$

$$\tilde{S}_p(f) = \{s \in C^1(\mathbb{R}^+, \mathbb{R}^n) ; \lim_{t \rightarrow \infty} s(t) = 0 \text{ and (58) holds at every } t > 0, z \in \mathbb{R}^n\}.$$

A natural candidate, besides our p -Santaló points $s_p(f_t)$ themselves, for which both terms are zero, would be a curve satisfying the equation

$$s'(t) = -\frac{p}{2q} \nabla_z Q(t, s(t)) = \frac{p}{2} \int_{\mathbb{R}^n} x \frac{\mathcal{L}_p(\tau_{s(t)} f_t)(x)}{\int_{\mathbb{R}^n} \mathcal{L}_p(\tau_{s(t)} f_t)(x) dx} dx. \quad (59)$$

We leave open the following natural question: does this time-dependent gradient flow equation have at least one (smooth enough) solution on $\mathbb{R}^+$ with $\lim_{t \rightarrow \infty} s(t) = 0$? In the case $p = 0$, the equation becomes

$$s'(t) = \frac{1}{2} \nabla \log L((f_t)^\square)(s(t)).$$

In all cases, one probably needs to understand the time dependent log-Laplace transform $Q(t, z)$.

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Figure 1: Open question on Santaló gradient-flow curves, source PDF page 33.